

A Sharp Endpoint-Denominator Theorem for Residue-Class Recurrences

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Abstract

We determine the exact uniform denominator of a family of inhomogeneous recurrences which decouple into residue classes modulo an arbitrary step q . For endpoint weight w , the residual modulus is

$$M(q, w) = \prod_{p|q} p^{\left\lceil \frac{w}{q} \left(v_p(q) + \frac{1}{p-1} \right) \right\rceil}.$$

The proof combines an elementary arithmetic-progression product bound with a unique-lowest-valuation endpoint argument. Binomial inversion transports the classification to the original companion basis. The parity-square theorem, whose sharp modulus is 4, is the case $(q, w) = (2, 2)$. A mixed step-three specialization proves L_n^2 -integrality, with optimal exponent, for the second solution of Zagier's sporadic sequence **B**; more generally its parameter family is integral in this sense exactly when 3 divides the parameter. At weight one the construction has an algebraic first row. Its cubic member gives geometrically convergent rational approximants to an explicit Fermat-cubic period, with the sharp denominator L_n .

1 The endpoint sequence

Write

$$L_n = \text{lcm}(1, 2, \dots, n), \quad L_0 = 1.$$

Fix integers $q \geq 2$, $w \geq 1$, and $b \in \mathbb{Z}$. For every prime $p \mid q$ put

$$\mu_p(q, w) = \left\lceil \frac{w}{q} \left(v_p(q) + \frac{1}{p-1} \right) \right\rceil, \quad M(q, w) = \prod_{p|q} p^{\mu_p(q, w)}. \quad (1)$$

Let $T_m = 0$ for $m \leq 0$, and define $T_n \in \mathbb{Q}$ for $n \geq 1$ by

$$n^w T_n = b^q (n - q + 1)^w T_{n-q} + (-b)^{n-1}. \quad (2)$$

Here $(-b)^0 = 1$, including when $b = 0$. Equivalently, for $T(z) = \sum_{n \geq 0} T_n z^n$ and $\theta = z d/dz$, recurrence (2) is the endpoint operator equation

$$\{\theta^w - b^q z^q (\theta + 1)^w\} T(z) = \frac{z}{1 + bz}. \quad (3)$$

For $0 \leq j < n$ such that $n - j \equiv 1 \pmod{q}$, set

$$Q_q(n, j) = \frac{\prod_{t=0}^{r-1} (j + 2 + tq)}{\prod_{t=0}^r (j + 1 + tq)}, \quad r = \frac{n - j - 1}{q}. \quad (4)$$

An empty numerator product is 1.

Theorem 1 (sharp step- q endpoint denominator). *For every $n \geq 1$,*

$$T_n = b^{n-1} \sum_{\substack{0 \leq j < n \\ n-j \equiv 1 \pmod{q}}} (-1)^j Q_q(n, j)^w. \quad (5)$$

Moreover, the following are equivalent:

1. $M(q, w) \mid b$;
2. $L_n^w b^{n-1} Q_q(n, j)^w \in \mathbb{Z}$ for every admissible (n, j) ;
3. $L_n^w T_n \in \mathbb{Z}$ for every $n \geq 0$.

The modulus in (1) is therefore both termwise sufficient and globally necessary. In particular, summation cannot lower it.

2 The finite endpoint expansion

Proof of (5). The forcing at index $j + 1$ contributes $(-b)^j / (j + 1)^w$. If $n = j + 1 + rq$, its propagation to n multiplies this by

$$\prod_{t=0}^{r-1} \frac{b^q (j + 2 + tq)^w}{(j + 1 + (t + 1)q)^w}.$$

The total power of b is $j + rq = n - 1$, and the remaining quotient is $Q_q(n, j)^w$. Summing over the admissible source indices proves the formula. $\square$

3 The arithmetic-progression bound

Lemma 2. *Let (n, j) be admissible and put $r = (n - j - 1)/q$.*

1. *If $p \nmid q$, then*

$$v_p(Q_q(n, j)) \geq -v_p(L_n). \quad (6)$$

2. *If $p \mid q$ and $s = v_p(q)$, then*

$$v_p(Q_q(n, j)) \geq -v_p(L_n) - rs - v_p(r!). \quad (7)$$

Proof. Suppose first that $p \nmid q$. For any fixed p^a , the denominator of $Q_q(n, j)$ contains $r + 1$ terms in one invertible residue progression, while the numerator contains r terms in another. The former residue occurs at most $\lceil (r + 1)/p^a \rceil$ times and the latter at least $\lfloor r/p^a \rfloor$ times. Their difference is at most one. Only powers $p^a \leq n$ occur, so summing over a proves (6).

Now suppose $p \mid q$. Write the denominator factors as $d_t = j + 1 + tq$, $0 \leq t \leq r$, and choose t_* so that $v_p(d_{t_*})$ is maximal. Since $d_{t_*} \leq n$, $v_p(d_{t_*}) \leq v_p(L_n)$. For $t \neq t_*$, maximality and $d_t - d_{t_*} = q(t - t_*)$ imply

$$v_p(d_t) \leq s + v_p(|t - t_*|).$$

Consequently

$$\begin{aligned} v_p \left(\prod_{t=0}^r d_t \right) &\leq v_p(L_n) + rs + v_p(t_*! (r - t_*)!) \\ &\leq v_p(L_n) + rs + v_p(r!), \end{aligned}$$

because $r!/[t_*! (r - t_*)!]$ is an integer. Discarding the numerator only weakens the estimate and proves (7). $\square$

4 Sufficiency and sharpness

Proof of Theorem 1. We have proved the finite formula. Assume first that $M(q, w) \mid b$. For $p \nmid q$, (6) makes $L_n^w Q_q(n, j)^w$ p -integral. For $p \mid q$, write $s = v_p(q)$ and $e = v_p(b)$. Definition (1) gives

$$eq \geq w \left(s + \frac{1}{p-1} \right). \quad (8)$$

Because $n-1 = j + rq \geq rq$ and $v_p(r!) \leq r/(p-1)$,

$$e(n-1) \geq eq \geq wr \left(s + \frac{1}{p-1} \right) \geq w(rs + v_p(r!)).$$

Together with (7), this proves the termwise assertion, and (5) proves $L_n^w T_n \in \mathbb{Z}$.

It remains to prove necessity. The case $b = 0$ is immediate, so assume $b \neq 0$. Suppose that $p \mid q$ and $e = v_p(b) < \mu_p(q, w)$; put $s = v_p(q)$. At $n = qR$, the admissible source indices are $j = q\ell - 1$, $1 \leq \ell \leq R$. Every numerator factor of $Q_q(qR, q\ell - 1)$ is 1 modulo p , while its denominator is

$$q^{R-\ell+1} \frac{R!}{(\ell-1)!}.$$

The valuation of this contribution to (5) is therefore

$$e(qR-1) - w \left((R-\ell+1)s + v_p \left(\frac{R!}{(\ell-1)!} \right) \right). \quad (9)$$

The $\ell = 1$ term is uniquely of least valuation: the difference between (9) and its value at $\ell = 1$ is

$$w((\ell-1)s + v_p((\ell-1)!)) > 0 \quad (\ell > 1).$$

Hence no cancellation is possible and

$$v_p(T_{qR}) = e(qR-1) - w(Rs + v_p(R!)). \quad (10)$$

Adding $wv_p(L_{qR})$, dividing by R , and letting $R \rightarrow \infty$ gives the limit

$$eq - w \left(s + \frac{1}{p-1} \right) < 0.$$

Thus $L_{qR}^w T_{qR}$ is nonintegral for all sufficiently large R . Condition (3) in the theorem forces $v_p(b) \geq \mu_p(q, w)$ for every $p \mid q$, which is exactly $M(q, w) \mid b$. $\square$

5 Binomial transport

Corollary 3. *Define*

$$B_n = \sum_{k=0}^n \binom{n}{k} b^{n-k} T_k. \quad (11)$$

Then

$$\boxed{L_n^w B_n \in \mathbb{Z} \text{ for every } n \iff M(q, w) \mid b.}$$

Proof. Binomial inversion gives

$$T_n = \sum_{k=0}^n \binom{n}{k} (-b)^{n-k} B_k.$$

Since $L_k \mid L_n$ for $k \leq n$, either transform preserves uniform L_n^w -integrality. Apply Theorem 1. $\square$

At the generating-function level, (11) and its inverse are exactly

$$B(z) = \frac{1}{1-bz} T\left(\frac{z}{1-bz}\right), \quad T(z) = \frac{1}{1+bz} B\left(\frac{z}{1+bz}\right).$$

Thus Corollary 3 transports the sharp arithmetic result from the decoupled operator (3) back to its binomially conjugate companion.

For $(q, w) = (2, 2)$, one obtains $M(2, 2) = 4$, recovering the parity-square endpoint theorem. If $q = \ell$ is prime, the answer simplifies markedly:

$$M(\ell, w) = \ell^{\lceil w/(\ell-1) \rceil}. \quad (12)$$

Thus $M(3, 2) = 3$ and $M(5, 4) = 5$, rather than the naive sufficient moduli 3^2 and 5^4 . The parity step is exceptional in that $M(2, w) = 2^w$.

q	w	$M(q, w)$
2	1, ..., 8	2, 4, 8, 16, 32, 64, 128, 256
3	1, ..., 8	3, 3, 9, 9, 27, 27, 81, 81
4	1, ..., 8	2, 4, 8, 8, 16, 32, 64, 64
5	1, ..., 8	5, 5, 5, 5, 25, 25, 25, 25
6	1, ..., 8	6, 6, 6, 12, 36, 36, 72, 72

6 A natural realization: Zagier's sporadic sequence B

The pure-power endpoint recurrence of Theorem 1 is not the only normal form with the same valuation density. A binomial conjugation of Zagier's quadratic recurrence [4] produces a mixed step-three endpoint in which the two missing residue classes occur once and the denominator class occurs twice.

For $h \in \mathbb{Z}$, let $C_0^{(h)} = 0$, $C_1^{(h)} = 1$, and

$$(n+1)^2 C_{n+1}^{(h)} = h(3n^2 + 3n + 1)C_n^{(h)} - 3h^2 n^2 C_{n-1}^{(h)}. \quad (13)$$

Define

$$S_n^{(h)} = \sum_{k=0}^n \binom{n}{k} (-h)^{n-k} C_k^{(h)}. \quad (14)$$

Theorem 4 (sharp step-three companion theorem). *For every integer h ,*

$$\left(\forall n \geq 0, L_n^2 C_n^{(h)} \in \mathbb{Z} \right) \iff 3 \mid h. \quad (15)$$

If $h = 3$, then (13) is the recurrence for the second solution of Zagier's sporadic sequence B. Consequently its companion satisfies

$$L_n^2 C_n^{(3)} \in \mathbb{Z} \quad (n \geq 0),$$

and the exponent 2 is optimal.

Proof. Let $C(x) = \sum_{n \geq 0} C_n^{(h)} x^n$. Including the initial residual, (13) is equivalent to

$$\{\theta^2 - hx(3\theta^2 + 3\theta + 1) + 3h^2 x^2(\theta + 1)^2\}C(x) = x.$$

The generating function of (14) is

$$S(z) = \frac{1}{1+hz} C\left(\frac{z}{1+hz}\right).$$

Direct conjugation cancels the terms of degrees one and two and gives

$$\{\theta^2 + h^3 z^3 (\theta + 1)(\theta + 2)\} S(z) = \frac{z}{1+hz}. \quad (16)$$

Thus

$$n^2 S_n^{(h)} + h^3 (n-2)(n-1) S_{n-3}^{(h)} = (-h)^{n-1}. \quad (17)$$

For $0 \leq j < n$ with $n-j \equiv 1 \pmod{3}$, put

$$r = \frac{n-j-1}{3}, \quad R(n, j) = \frac{\prod_{t=0}^{r-1} (j+2+3t)(j+3+3t)}{\prod_{t=0}^r (j+1+3t)^2}.$$

Iteration of (17) yields

$$(-1)^{n-1} S_n^{(h)} = h^{n-1} \sum_{\substack{0 \leq j < n \\ n-j \equiv 1 \pmod{3}}} R(n, j). \quad (18)$$

Indeed, a source at $j+1$ acquires r propagation signs, and $j+r \equiv n-1 \pmod{2}$. In particular, for $h > 0$ the right-hand side of (18) is a sum of positive rational numbers.

We prove termwise integrality when $3 \mid h$. If $p \neq 3$, then for every power p^a the doubled denominator progression contains at most $2\lceil (r+1)/p^a \rceil$ multiples of p^a , while the two numerator progressions contain at least $2\lfloor r/p^a \rfloor$ in total. The deficit is at most two. Hence

$$v_p(R(n, j)) \geq -2v_p(L_n). \quad (19)$$

For $p = 3$, apply the maximal-factor argument of Lemma 2 to $D = \prod_{t=0}^r (j+1+3t)$. It gives

$$v_3(D) \leq v_3(L_n) + r + v_3(r!),$$

and therefore

$$v_3(R(n, j)) \geq -2v_3(L_n) - 2r - 2v_3(r!). \quad (20)$$

If $3 \mid h$, then $n-1 = j+3r \geq 3r$ and $v_3(r!) \leq r/2$, so

$$v_3(L_n^2 h^{n-1} R(n, j)) \geq 3r - 2r - 2v_3(r!) \geq 0.$$

Together with (19), this proves every summand of (18) integral after multiplication by L_n^2 . Binomial inversion and $L_k \mid L_n$ transfer the result to $C_n^{(h)}$.

For necessity, exact propagation of (13) gives

$$C_6^{(h)} = -\frac{39521h^5}{32400}, \quad L_6^2 C_6^{(h)} = -\frac{39521h^5}{9}. \quad (21)$$

Since $3 \nmid 39521$, integrality at $n = 6$ forces $3 \mid h$. This proves (15).

For $h = 3$, the first solution of (13) is the named sporadic row [4, 2]

$$A_{\mathbf{B}}(n) = \sum_{k=0}^{\lfloor n/3 \rfloor} (-1)^k 3^{n-3k} \binom{n}{3k} \frac{(3k)!}{k!^3},$$

with $A_{\mathbf{B}}(0) = 1$, $A_{\mathbf{B}}(1) = 3$. Hence $C^{(3)}$ is its normalized companion. Finally, $C_2^{(3)} = 21/4$, so $L_2 C_2^{(3)} = 21/2 \notin \mathbb{Z}$; the exponent two cannot be lowered. $\square$

7 A cubic algebraic period with sharp linear denominators

The weight-one specialization has an additional structure not visible in the general valuation statement: its first row is algebraic. We record the cubic case in detail because it turns the endpoint theorem into a new Apéry-type approximation scheme. Four-term Apéry-like recurrences have been studied systematically by Cooper [1]; we have not located the algebraic row below, its distinguished companion, or the sharp companion-denominator statement in that literature.

Let $b \in \mathbb{Z}$ and put

$$Q_b(x) = (1 - bx)^3 - (bx)^3 = 1 - 3bx + 3b^2x^2 - 2b^3x^3. \quad (22)$$

Define $A^{(b)}(x) = \sum_{n \geq 0} A_n^{(b)} x^n$ and $C^{(b)}(x) = \sum_{n \geq 0} C_n^{(b)} x^n$ by

$$A^{(b)}(x) = Q_b(x)^{-1/3}, \quad (23)$$

$$Q_b(x) \frac{d}{dx} C^{(b)}(x) + \frac{1}{3} Q'_b(x) C^{(b)}(x) = 1 - bx, \quad C^{(b)}(0) = 0. \quad (24)$$

Theorem 5 (cubic algebraic-period companion). *The two rows satisfy, for $n \geq 2$, the common four-term recurrence*

$$(n+1)Y_{n+1} = b(3n+1)Y_n - b^2(3n-1)Y_{n-1} + 2b^3(n-1)Y_{n-2}. \quad (25)$$

Their initial values are

$$(A_0^{(b)}, A_1^{(b)}, A_2^{(b)}) = (1, b, b^2), \quad (C_0^{(b)}, C_1^{(b)}, C_2^{(b)}) = (0, 1, 3b/2).$$

Moreover,

$$(\forall n \geq 0, A_n^{(b)} \in \mathbb{Z}) \iff 3 \mid b, \quad (26)$$

$$(\forall n \geq 0, L_n C_n^{(b)} \in \mathbb{Z}) \iff 3 \mid b. \quad (27)$$

For $b > 0$ the companion ratios converge to the cubic period

$$\lim_{n \rightarrow \infty} \frac{C_n^{(b)}}{A_n^{(b)}} = \frac{\Lambda_3}{b}, \quad \Lambda_3 = \int_0^1 \frac{dt}{(1+t)(1-t^3)^{2/3}}. \quad (28)$$

The convergence is geometric; more precisely its exponential factor is at most $1/2$.

Proof. First take the homogeneous endpoint row $U_0 = 1, U_1 = U_2 = 0$ and

$$nU_n = b^3(n-2)U_{n-3}.$$

Its generating function is

$$U(z) = \{1 - (bz)^3\}^{-1/3}, \quad U_{3r} = b^{3r} \frac{(1/3)_r}{r!}. \quad (29)$$

Let T be the endpoint companion of (2) for $(q, w) = (3, 1)$, and transport both rows back through the inverse binomial transform:

$$A_n^{(b)} = \sum_{k=0}^n \binom{n}{k} b^{n-k} U_k, \quad C_n^{(b)} = \sum_{k=0}^n \binom{n}{k} b^{n-k} T_k. \quad (30)$$

At the generating-function level this is

$$A^{(b)}(x) = \frac{1}{1-bx} U\left(\frac{x}{1-bx}\right), \quad C^{(b)}(x) = \frac{1}{1-bx} T\left(\frac{x}{1-bx}\right).$$

The first identity immediately gives (23). Conjugating the first-order endpoint equation gives (24). Equating coefficients in (23)–(24) proves (25) and the stated initial values.

For completeness, the first row also has the finite form

$$A_n^{(b)} = b^n \sum_{0 \leq r \leq n/3} \binom{n}{3r} \frac{(1/3)_r}{r!}. \quad (31)$$

If $b = 3h$, every summand is integral. Indeed it is enough to take $b = 3$ and $n = 3r$. Writing $P_r = \prod_{t=0}^{r-1} (3t+1)$, the residual factor is

$$3^{3r} \frac{(1/3)_r}{r!} = 3^{2r} \frac{P_r}{r!}.$$

For every prime $p \neq 3$ and every $a \geq 1$, the progression $1, 4, \dots, 3r-2$ contains at least $\lfloor r/p^a \rfloor$ multiples of p^a , because multiplication by 3 permutes the residue classes modulo p^a . Thus $v_p(P_r) \geq v_p(r!)$. At $p = 3$ the factor 3^{2r} is more than sufficient. This proves the forward implication of (26); the reverse implication follows already from

$$A_3^{(b)} = \frac{4b^3}{3}.$$

For the companion, formula (1) gives $M(3, 1) = 3$. Theorem 1 and the binomial transport (30) therefore prove $L_n C_n^{(b)} \in \mathbb{Z}$ whenever $3 \mid b$. Sharpness is visible at one finite index:

$$C_6^{(b)} = \frac{1507b^5}{180}, \quad L_6 C_6^{(b)} = \frac{1507b^5}{3}. \quad (32)$$

Since $3 \nmid 1507$, integrality at $n = 6$ forces $3 \mid b$. This proves (27) without an asymptotic necessity argument.

It remains to identify the limit. Assume $b \neq 0$, put $s = bz$, and write $T(z) = b^{-1}\tau(s)$. The endpoint differential equation becomes

$$(1-s^3)\tau'(s) - s^2\tau(s) = \frac{1}{1+s}.$$

Variation of constants gives the exact solution

$$\tau(s) = (1-s^3)^{-1/3} \int_0^s \frac{dt}{(1+t)(1-t^3)^{2/3}}. \quad (33)$$

Consequently, with $s = bx/(1-bx)$,

$$\frac{C^{(b)}(x)}{A^{(b)}(x)} = \frac{1}{b} \int_0^s \frac{dt}{(1+t)(1-t^3)^{2/3}}. \quad (34)$$

The unique dominant singularity for $b > 0$ is $s = 1$, corresponding to $x = 1/(2b)$. The other two cubic roots of unity correspond to points of modulus $1/b$. Near $s = 1$, the difference between the integral in (34) and Λ_3 is $(1-s)^{1/3}$ times an analytic unit; multiplication by $(1-s^3)^{-1/3}$ removes this local branch. Standard coefficient transfer therefore gives (28), while the next singularity radius gives exponential factor at most $(1/(2b))/(1/b) = 1/2$. $\square$

Corollary 6 (the explicit cubic row). *For $b = 3$ the integral first row begins*

$$1, 3, 9, 36, 189, 1053, 5751, 31104, 169857, \dots,$$

and its distinguished companion begins

$$0, 1, \frac{9}{2}, \frac{33}{2}, \frac{279}{4}, \frac{7263}{20}, \frac{40689}{20}, \dots$$

They satisfy

$$(n+1)Y_{n+1} = 3(3n+1)Y_n - 9(3n-1)Y_{n-1} + 54(n-1)Y_{n-2},$$

and

$$L_n C_n^{(3)} \in \mathbb{Z} \quad (n \geq 0), \quad \frac{C_n^{(3)}}{A_n^{(3)}} \longrightarrow 0.3643879210972249599026202027563627900 \dots$$

The exponent one is optimal, since $C_2^{(3)} = 9/2 \notin \mathbb{Z}$.

8 Interpretation and reproducibility

The exponent in (1) is a density balance. The contribution $v_p(q)$ is built into every step of the denominator progression, while $1/(p-1)$ is the asymptotic valuation density of a factorial. The factor $1/q$ records that only one in every q indices lies in a given endpoint chain. The unique minimum in (10) proves that this density cannot be improved by cancellations between endpoints.

The closest classical arithmetic-progression literature studies least common multiples or products of consecutive progression terms; see, for example, Hong and Qian [3]. The present statement instead determines the sharp parameter modulus for a ratio of interlaced progressions arising from an inhomogeneous recurrence.

The exact audit

`work/harmonic_jets/verify_step_q_endpoint.py`

checks the recurrence against (5), the termwise bound over a grid of (q, w, n) , and deficient-prime witnesses for the necessity argument. The independent application audit

`work/harmonic_jets/verify_zagier_B_endpoint.py`

checks the operator conjugation for arbitrary h , the positive endpoint formula, termwise integrality, the named binomial first row, and the two exact sharpness witnesses in Theorem 4. The cubic-period audit

`work/harmonic_jets/verify_cubic_period.py`

checks the algebraic conjugation, the common four-term recurrence, both finite sharpness witnesses, exact integrality through $n = 180$, and the period limit to high precision.

References

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